2. Technical Project Report

2.1 Problem Statement

The rapid advancement of generative artificial intelligence has significantly increased the creation of realistic synthetic media. Deepfakes can be used for misinformation, impersonation, reputation damage, fraud, and social engineering attacks. Existing users often lack accessible tools to verify whether digital media is authentic.

The objective of DeepShield is to investigate practical methods for detecting manipulated media and provide understandable analysis results through a user-friendly platform.

2.2 Objectives

The project was designed with the following objectives:

1. Detect manipulated images using deep learning models.
2. Extend detection capabilities to videos.
3. Provide explainable AI outputs using Grad-CAM visualizations.
4. Build a mobile-first interface for accessibility.
5. Develop a scalable backend architecture for inference.
6. Explore the cybersecurity implications of synthetic media attacks.
7. Investigate future multimodal verification approaches involving text and news content.

2.3 System Architecture

The system follows a client-server architecture.

Mobile Application

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FastAPI Backend

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Inference Engine

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Deep Learning Models

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Prediction + Explainability

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Result Visualization

Workflow:

1. User uploads image or video.
2. Media is transmitted securely to the backend.
3. Deep learning model performs inference.
4. Confidence score is generated.
5. Grad-CAM explanation map is produced.
6. Results are returned to the mobile application.

2.4 Technology Stack

Frontend

- Flutter
- Dart
- Image Picker
- Shared Preferences

Backend

- Python
- FastAPI
- Uvicorn
- SQLAlchemy
- SQLite

Machine Learning

- PyTorch
- EfficientNet
- ResNet
- Grad-CAM

Infrastructure

- AWS (evaluation and experimentation)

- Android devices via ADB testing

Development Tools

- Git
- Postman
- VS Code
- Linux (Kali Linux)

2.5 Core Features

Image Deepfake Detection

Users can upload images for authenticity analysis.

Video Analysis

Video files are processed through frame extraction and aggregated predictions.

Explainable AI

Grad-CAM heatmaps highlight image regions influencing model decisions.

Confidence Scoring

The system provides prediction confidence values alongside labels.

Mobile Interface

A lightweight Flutter application enables on-device interaction with the backend.

API-Based Design

Inference functionality is exposed through REST endpoints.

2.6 Challenges Encountered

Dataset Quality

Obtaining balanced and representative datasets remains challenging.

Model Generalization

Models often perform differently on unseen data and across datasets.

Explainability Integration

Generating meaningful visual explanations required experimentation with Grad-CAM configurations.

Mobile-Backend Communication

Reliable communication between Flutter applications and backend services required debugging networking and deployment configurations.

Resource Constraints

Training large models requires significant computational resources.

2.7 Solutions Implemented

Data Processing Pipeline

Developed scripts for image extraction, preprocessing, and frame generation.

Explainable Inference

Integrated Grad-CAM to improve transparency.

Modular Backend Design

Separated inference, API, database, and utility components.

Mobile Integration

Implemented Flutter-based upload and result workflows.

Video Aggregation

Used frame-level analysis and majority voting techniques for video classification.

2.8 Current Status

Completed:

- FastAPI backend
- Image upload workflow
- Deepfake image inference
- Video inference endpoint
- Grad-CAM generation
- Flutter application recovery and integration
- End-to-end prediction pipeline
- Result visualization

In Progress:

- Improved model accuracy
- Enhanced explainability

- Video user interface
- Research validation

2.9 Future Work

Multimodal Analysis

Combine image, video, and textual information.

News Verification

Analyze media alongside associated news content.

Improved Model Architectures

Evaluate larger EfficientNet variants and transformer-based approaches.

Indian Context Dataset

Develop datasets representing regional and demographic diversity.

Media Authenticity Platform

Expand DeepShield into a broader cybersecurity platform for misinformation detection and digital trust assessment.